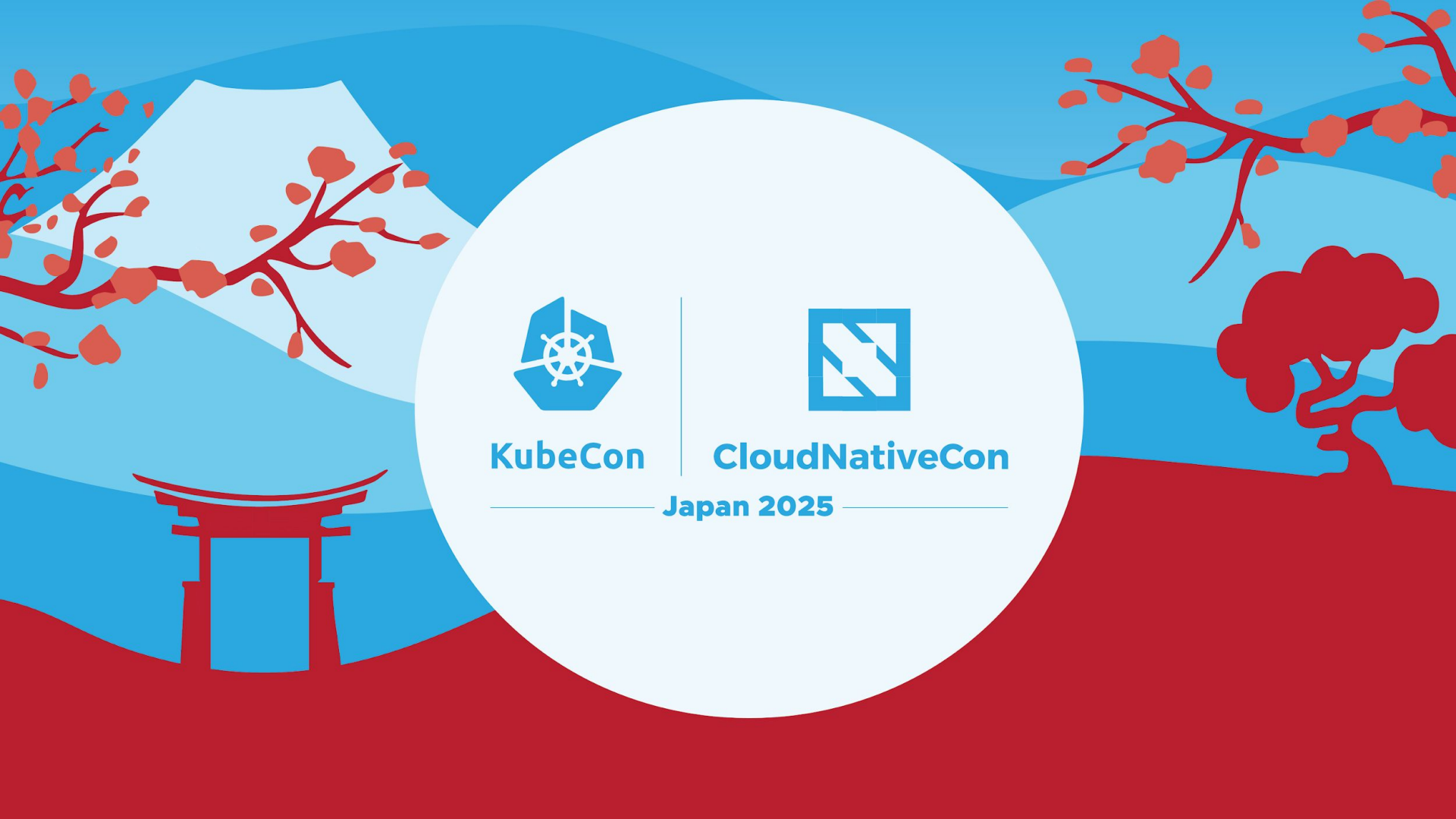
KubeCon
CloudNativeCon
Japan 2025

Kubernetes SIG Node Intro and Deep Dive

Paco Xu, DaoCloud
Narang Dixita Sohanlal, Google
Hironori Shiina, Independent

Content

Agenda

01 SIG Node Intro

02 Feature updates in 1.33 and 1.34

03 Highlight:

- *DRA*
- *In-place Pod Resize*

04 Get involved

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01 SIG Node Intro

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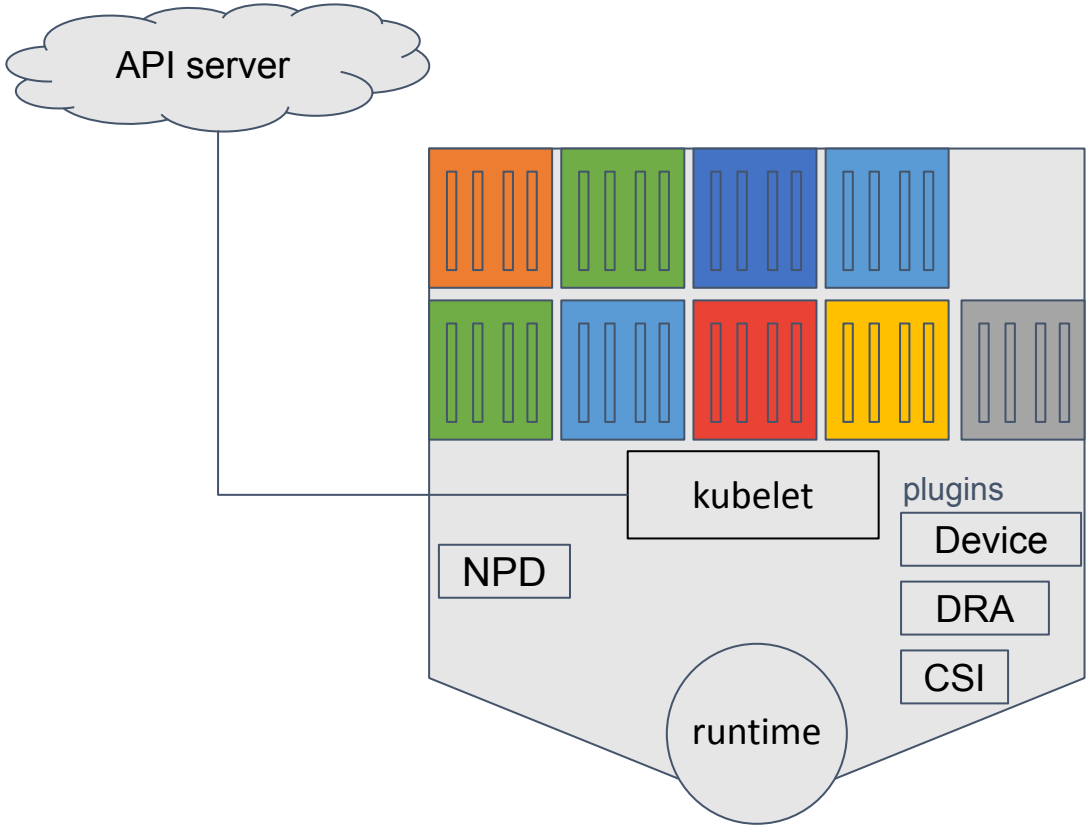
SIG Node Components



SIG Node Components

- **Kubelet**: The foreman on the node
- **Container Runtime**: The Engine Room
- **Device plugins / DRA**: Assigning the hardware
- **Node Problem Detector**: The Watchdog
- **Storage and Network Components**: CSI, CNI

Together, a Symphony of functionality

- Run containerized workloads efficiently
- Manage pod lifecycles
- Preserve node health, enabling self-heal
- Facilitate smooth communication and integration with storage resources
- Manage resources
- Optimize the use of specialized hardware resource

Resource Management: The Key to Efficient Workloads

kubelet source tree

kubernetes / pkg / kubelet /

- allocation
- apis
- cadvisor
- certificate
- checkpointmanager
- client
- clustertrustbundle
- cm
- config
- configmap
- container
- envvars
- events
- eviction
- images
- kubeletconfig
- kuberuntime

kubernetes / pkg / kubelet /

- lifecycle
- logs
- metrics
- network
- nodeshutdown
- nodestatus
- oom
- pleg
- pluginmanager
- pod
- preemption
- prober
- qos
- runtimeclass
- secret
- server
- stats

kubernetes / pkg / kubelet /

- status
- sysctl
- token
- types
- userns
- util
- volumemanager
- watchdog
- winstats
- OWNERS
- active_deadline.go
- active_deadline_test.go
- doc.go
- errors.go
- kubelet.go
- kubelet_getters.go

kubernetes / pkg / kubelet /

- kubelet_linux.go
- kubelet_network.go
- kubelet_network_linux.go
- kubelet_network_others.go
- kubelet_node_status.go
- kubelet_node_status_others.go
- kubelet_node_status_test.go
- kubelet_node_status_windows.go
- kubelet_others.go
- kubelet_pods.go
- kubelet_pods_linux_test.go
- kubelet_pods_test.go
- kubelet_pods_windows_test.go
- kubelet_resources.go
- kubelet_resources_test.go
- kubelet_server_journal.go
- kubelet_server_journal_linux.go

kubelet: Container Lifecycle Management

- **config:** Watch pods in API to detect addition, update and deletion
- **lifecycle:** Check if pod can be admitted to node
- **pod:** Store desired pod specs
- **kuberuntime, container:** Handle pods and containers with the runtime
  - Create and stop pod
  - Create, start and stop containers in the appropriate order
    (init containers / sidecar containers, regular containers, ephemeral containers)
  - Restart container based on its restart policy
  - Resize containers
  - Run post-start and pre-stop hooks
- **pleg:** Detect container status change in the runtime
- **status:** Report pod status to API server
- **prober:** Run probes (liveness, readiness, startup)
- **volumemanager:** Attach volumes to containers
- **image:** Pull and manage container images
- **logs:** Rotate log files of containers

kubelet: Resource Management

- **cm:** Container Manager
  - Configure cgroups for pod QoS (CPU, memory, huge page)
  - **cpumanager:** Assign CPUs to containers based on policy
  - **memorymanager:** Assign memory nodes to containers based on policy
  - **topologymanager:** Optimize resource alignment
  - **devicemanager:** Attach devices to containers
  - **dra:** Allocate devices to containers dynamically
- **eviction:** Evict pods that consume too much resource
  (memory, ephemeral storage, PID, inode)

kubelet: others

- **kubeletconfig:** kubelet configuration
- **nodestatus:** Report node status to API server
- **nodeshutdown:** Detect node shutdown and terminate pods
- **configmap, secret:** Cache configmap and secret to be attached to container
- **runtimeclass:** Store runtime class information
- **userns:** Store ID mappings for user namespace feature

And more

Content

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04 Get involved

Major Feature Updates: Pod Lifecycle

In-Place Pod Resize: Graduated to Beta

```yaml
apiVersion: v1
kind: Pod
spec:
  containers:
  - resizePolicy:
    - resourceName: cpu
      restartPolicy: NotRequired
    - resourceName: memory
      restartPolicy: NotRequired
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
```

```yaml
apiVersion: v1
kind: Pod
spec:
  containers:
  - resizePolicy:
    - resourceName: cpu
      restartPolicy: NotRequired
    - resourceName: memory
      restartPolicy: NotRequired
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
```

Major Feature Updates: Pod Lifecycle

Sidecar Containers: Graduated to GA

```yaml
spec:
  containers:
    - name: myapp
      image: alpine:latest
      command: ['sh', '-c', 'while true; do echo "logging" >> /opt/logs.txt; sleep 1; done']
      volumeMounts:
        - name: data
          mountPath: /opt
  initContainers:
    - name: logshipper
      image: alpine:latest
      restartPolicy: Always
      command: ['sh', '-c', 'tail -F /opt/logs.txt']
      volumeMounts:
        - name: data
          mountPath: /opt
  volumes:
    - name: data
      emptyDir: {}
```

Pod Spec Option Updates

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod
spec:
  hostUsers: true # 1
  containers:
  - image: docker.io/library/alpine:latest
    name: ctr
    procMountType: Unmasked # 2

    lifecycle:
      stopSignal: SIGHUP # 3
      preStop: # 4
        sleep: # 4
          seconds: 0 # 5

    volumeMounts: # 6
    - name: volume # 6
      mountPath: /volume # 6
  volumes:
  - name: volume # 6
    image: # 6
      reference: quay.io/crio/artifact:v1 # 6
      pullPolicy: IfNotPresent # 6
  securityContext:
    supplementalGroupsPolicy: Strict # 7
```

1. User Namespaces in pods on by default beta
2. add ProcMount option on by default beta
3. Container Stop Signals alpha
4. sleep action for PreStop hook beta
5. zero value for PreStop sleep action beta
6. OCI Volume Mount beta
7. Fine-grained supplemental groups beta

Kubelet Configuration Option Updates


```yaml
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
imagePullCredentialsVerificationPolicy: AlwaysVerify # 1
crashLoopBackOff: # 2
    maxContainerRestartPeriod: "100s" # 2


apiVersion: kubelet.config.k8s.io/v1
kind: CredentialProviderConfig
providers:
- name: token
  tokenAttributes: # 3
   serviceAccountTokenAudience: test-audience # 3
   requireServiceAccount: true # 3
   requiredServiceAccountAnnotationKeys: # 3
   - "domain.io/identity-id" # 3
   - "domain.io/identity-type" # 3
```

1. Ensure secret pulled images alpha
2. Tune crash loop backoff alpha
3. Projected service account tokens for image credential alpha

Miscellaneous

- Pressure Stall Information to alpha

Three CPU manager policies have made progress:

- Split L3 cache to alpha 2
- Option to distribute across NUMA nodes to beta
- reject non SMT-aligned workloads went GA

Enhancement for V1.34

Now Planning

Candidates:

- DRA: structured parameters
- VolumeSource: OCI Artifact and/or Image
- In-Place Update of Pod Resources
- DRA: Handle extended resource requests via DRA Driver
- DRA: Consumable Capacity
- Pod Generation
- DRA: Resource Claim Status with possible standardized network interface data
- DRA: Partitionable Devices
- Add Resource Health Status to the Pod Status for Device Plugin and DRA
- Pod level resources
- Node memory swap support
- Windows node gracefully shutdown

And more

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DRA (Dynamic Resource Allocation)

Flexible device allocation for pods

- Share devices between containers
- Share devices between pods
- Pass parameters to devices

Graduated to Beta at V1.32

Updates in V1.33:

- Partitionable Devices (alpha)
- Device taints and tolerations (alpha)
- Admin Access (alpha)

More features are being developed actively for the future releases

DRA: Allocate one device to each pod

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test1
  name: pod0
spec:
  containers:
  - name: ctr0
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: gpu
  resourceClaims:
  - name: gpu
    resourceClaimTemplateName: single-gpu
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test1
  name: pod1
spec:
  containers:
  - name: ctr0
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: gpu
  resourceClaims:
  - name: gpu
    resourceClaimTemplateName: single-gpu
```

```yaml
apiVersion: resource.k8s.io/v1beta1
kind: ResourceClaimTemplate
metadata:
  namespace: gpu-test1
  name: single-gpu
spec:
  spec:
    devices:
      requests:
      - name: gpu
        deviceClassName: gpu.example.com
```

https://github.com/kubernetes-sigs/dra-example-driver/blob/main/demo/gpu-test1.yaml

DRA: Allocate one device to each pod


```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test1
  name: pod0
spec:
  :
  resourceClaims:
  - name: gpu
    resourceClaimTemplateName: single-gpu
  :
status:
  :
  resourceClaimStatuses:
  - name: gpu
    resourceClaimName: pod0-gpu-9r6vj
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test1
  name: pod1
spec:
  :
  resourceClaims:
  - name: gpu
    resourceClaimTemplateName: single-gpu
  :
status:
  :
  resourceClaimStatuses:
  - name: gpu
    resourceClaimName: pod1-gpu-mwfcc
```

```yaml
apiVersion: resource.k8s.io/v1beta1
kind: ResourceClaim
metadata:
  name: pod0-gpu-9r6vj
  namespace: gpu-test1
spec:
  devices:
    requests:
    - allocationMode: ExactCount
      count: 1
      deviceClassName: gpu.example.com
      name: gpu
```

```yaml
apiVersion: resource.k8s.io/v1beta1
kind: ResourceClaim
metadata:
  name: pod1-gpu-mwfcc
  namespace: gpu-test1
spec:
  devices:
    requests:
    - allocationMode: ExactCount
      count: 1
      deviceClassName: gpu.example.com
      name: gpu
```

DRA: Share device between containers

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test3
  name: pod0
spec:
  containers:
  - name: ctr0
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpu
  - name: ctr1
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpu
  resourceClaims:
  - name: shared-gpu
    resourceClaimTemplateName: single-gpu
```

```yaml
apiVersion: resource.k8s.io/v1beta1
kind: ResourceClaimTemplate
metadata:
  namespace: gpu-test3
  name: single-gpu
spec:
  spec:
    devices:
      requests:
      - name: gpu
        deviceClassName: gpu.example.com
```

https://github.com/kubernetes-sigs/dra-example-driver/blob/main/demo/gpu-test3.yaml

DRA: Share device between pods

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test4
  name: pod0
spec:
  containers:
  - name: ctr0
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpu
  resourceClaims:
  - name: shared-gpu
    resourceClaimName: single-gpu
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test4
  name: pod1
spec:
  containers:
  - name: ctr0
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpu
  resourceClaims:
  - name: shared-gpu
    resourceClaimName: single-gpu
```

```yaml
apiVersion: resource.k8s.io/v1beta1
kind: ResourceClaim
metadata:
  namespace: gpu-test4
  name: single-gpu
spec:
  devices:
    requests:
    - name: gpu
      deviceClassName: gpu.example.com
```

https://github.com/kubernetes-sigs/dra-example-driver/blob/main/demo/gpu-test4.yaml

DRA: Configure devices

```yaml
apiVersion: resource.k8s.io/v1beta1
kind: ResourceClaimTemplate
metadata:
  namespace: gpu-test5
  name: multiple-gpus
spec:
  spec:
    devices:
      requests:
      - name: ts-gpu
        deviceClassName: gpu.example.com
      - name: sp-gpu
        deviceClassName: gpu.example.com
      config:
      - requests: ["ts-gpu"]
        opaque:
          driver: gpu.example.com
          parameters:
            apiVersion: gpu.resource.example.com/v1alpha1
            kind: GpuConfig
            sharing:
              strategy: TimeSlicing
              timeSlicingConfig:
                interval: Long
```

```yaml
      - requests: ["sp-gpu"]
        opaque:
          driver: gpu.example.com
          parameters:
            apiVersion: gpu.resource.example.com/v1alpha1
            kind: GpuConfig
            sharing:
              strategy: SpacePartitioning
              spacePartitioningConfig:
                partitionCount: 10
```

https://github.com/kubernetes-sigs/dra-example-driver/blob/main/demo/gpu-test5.yaml

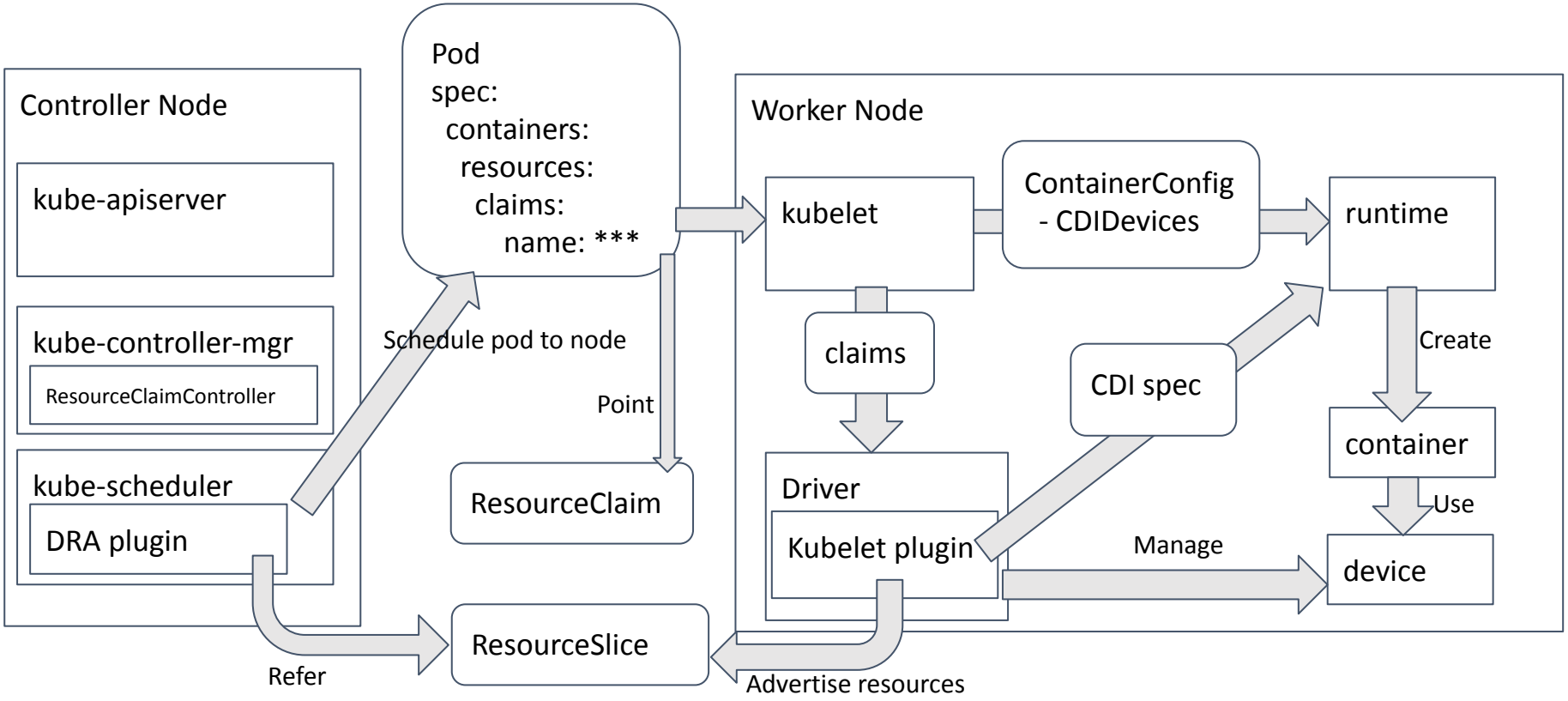
DRA: Configure devices

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: gpu-test5
  name: pod0
spec:
  containers:
  - name: ts-ctr0
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpus
        request: ts-gpu
  - name: ts-ctr1
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpus
        request: ts-gpu
```

```yaml
  - name: sp-ctr0
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpus
        request: sp-gpu
  - name: sp-ctr1
    image: ubuntu:22.04
    command: ["bash", "-c"]
    args: ["export; trap 'exit 0' TERM; sleep 9999 & wait"]
    resources:
      claims:
      - name: shared-gpus
        request: sp-gpu
  resourceClaims:
  - name: shared-gpus
    resourceClaimTemplateName: multiple-gpus
```

https://github.com/kubernetes-sigs/dra-example-driver/blob/main/demo/gpu-test5.yaml

DRA: Implementation overview



In-Place Pod Resize

Allow updating allocated resources(CPU and memory) to containers without restarting

```yaml
apiVersion: v1
kind: Pod
spec:
  containers:
  - resizePolicy:
    - resourceName: cpu
      restartPolicy: NotRequired
    - resourceName: memory
      restartPolicy: NotRequired
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
```

```yaml
apiVersion: v1
kind: Pod
spec:
  containers:
  - resizePolicy:
    - resourceName: cpu
      restartPolicy: NotRequired
    - resourceName: memory
      restartPolicy: NotRequired
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
```

In-Place Pod Resize

Graduated to Beta at V1.33

- Actual resources in container status are updated faster
- Sidecar containers are supported
- Many bugs have been fixed

Current Limitations:

- Decrease memory limits
- Resize CPUs with static policy
- Resize memory requests of a container with swap enabled
- Remove resource requests/limits

In-Place Pod Resize - v1.31 (alpha)

Resize this pod

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
:
status:
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
:
```

In-Place Pod Resize - v1.31 (alpha)

See cgroup values

```
$ kubectl exec resize-pod -- cat /sys/fs/cgroup/cpu.max
20000 100000
$ kubectl exec resize-pod -- cat /sys/fs/cgroup/cpu.weight
8
$ kubectl exec resize-pod -- cat /sys/fs/cgroup/memory.max
209715200
```

In-Place Pod Resize - v1.31 (alpha)

Request resize

```
$ kubectl patch pod resize-pod \
> --patch '{"spec":{"containers":[{"name":"resize-container", \
> "resources":{"requests":{"cpu":"300m", "memory":"300Mi"}, \
>              "limits":{"cpu":"300m", "memory": "300Mi"}}}]}}'
pod/resize-pod patched
```

In-Place Pod Resize - v1.31 (alpha)

Reize proposed

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
status:
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
:
  resize: Proposed
:
```

In-Place Pod Resize - v1.31 (alpha)

Get into InProgress

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
status:
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
:
  resize: InProgress
:
```

In-Place Pod Resize - v1.31 (alpha)

cgroup values have been updated soon

```
$ kubectl exec resize-pod -- cat /sys/fs/cgroup/cpu.max
30000 100000
$ kubectl exec resize-pod -- cat /sys/fs/cgroup/cpu.weight
12
$ kubectl exec resize-pod -- cat /sys/fs/cgroup/memory.max
314572800
```

In-Place Pod Resize - v1.31 (alpha)

But, still InProgress…

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
status:
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
:
  resize: InProgress
:
```

In-Place Pod Resize - v1.31 (alpha)

After 60 - 90 seconds

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
status:
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
```

In-Place Pod Resize - v1.33 (beta)

Let’s try again on V1.33

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
:
status:
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
        memory: 200Mi
:
```

In-Place Pod Resize - v1.33 (beta)

Request resize (via the ***resize*** subresource)

```
$ kubectl patch pod resize-pod --subresource=resize \
> --patch '{"spec":{"containers":[{"name":"resize-container", \
> "resources":{"requests":{"cpu":"300m", "memory":"300Mi"}, \
>              "limits":{"cpu":"300m", "memory": "300Mi"}}}]}}'
pod/resize-pod patched
```

In-Place Pod Resize - v1.33 (beta)

Get into InProgress
(via ***Condition***)

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
status:
  conditions:
  - lastProbeTime: "2025-05-21T14:41:53Z"
    lastTransitionTime: "2025-05-21T14:41:53Z"
    status: "True"
    type: PodResizeInProgress
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 200m
        memory: 200Mi
      requests:
        cpu: 200m
```

In-Place Pod Resize - v1.33 (beta)

Complete immediately

```
$ kubectl get pod resize-pod -o yaml
:
spec:
:
  containers:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
status:
:
  containerStatuses:
:
    resources:
      limits:
        cpu: 300m
        memory: 300Mi
      requests:
        cpu: 300m
        memory: 300Mi
:
```

Content

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01 SIG Node Intro

02 Feature updates in 1.33、1.34

03 Highlight:

- *DRA*
- *In-place Pod Resize*

04 Get involved

Get involved!

- KEP Wranglers
- CI subgroup
- Fix bugs
- PR review/feedback
- Documentation
- Try out features
- Attend SIG-Node
- Write features

GET INVOLVED!

KEP Wranglers

CI subgroup

Fix bugs

PR review/
feedback

Try out/ Write features

Attend SIG-Node